Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Multi Observation

Global Ocean Sea Surface Salinity, Sea Surface Density and Spiciness Product

MULTIOBS_GLO_PHY_SSS_L4_MY_015_015

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Change Record

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the external product **MULTIOBS_GLO_PHY_SSS_L4_MY_015_015**, developed by the Laboratory of Physical and Space Oceanography (LOPS, UMR6523 UBO/CNRS/Ifremer/IRD), which provides global Level 4 analyses of the Sea Surface Salinity (SSS), Sea Surface Density (SSD) and Sea Surface Spiciness (SSSp), along with Sea Surface Absolute Salinity (SSA), Conservative Temperature (SCT), surface thermal expansion coefficient (alpha) and haline contraction coefficient (beta). The SSS product is obtained using an optimal interpolation (OI) algorithm, that combines ISAS in situ SSS OI analyses, ESA Soil Moisture Ocean Salinity (SMOS) satellite and NASA Soil Moisture Active Passive images to reduce large scale and temporally varying bias. The SSS L4 product outcome is then combined with satellite SST products to compute thermodynamic sea water parameters using TEOS-10 (Kolodziejczyk et al., 2021).

It includes 1 dataset: *cmems_obs-mob_glo_phy-sss_my_multi-oi_P1W* which provides weekly data.

The **MULTIOBS_GLO_PHY_SSS_L4_MY_015_015** is produced at CATDS (Centre Aval de Traitement des Données). As input, the **MULTIOBS_GLO_PHY_SSS_L4_MY_015_015** uses monthly ISAS SSS analysis from in situ data from Coriolis data center, SSS SMOS L3Q mixed product from RE07 ESA L2 processing (http://dx.doi.org/10.12770/0f02fc28-cb86-4c44-89f3-ee7df6177e7b) and Remote Sensing System SMAP L3 v5.3 product. The SSS L4 analyses are combined with AMSR2 SST satellite data to compute SSD, SSSp, SSA, SCT and alpha/beta coefficients using TEOS-10 algorithm (Kolodziejczyk et al., 2021).

SSS, SSD, SSSp L4 products are provided on an EASE grid (0.25°), with a weakly temporal resolution and an effective resolution of about 125 km. ***Caution: the global EASE grid degrades the meridional resolution at high latitude.***

The NRT product MULTIOBS_GLO_PHY_SSS_L3_MYNRT_015_014 is used as a base to build the weekly input fields that are used to create this product (MULTIOBS_GLO_PHY_SSS_L3_MYNRT_015_015).

I.2 Summary of the results

MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 quality is estimated through comparison with independent in situ measurements collected by thermosalinographs from merchant ships (TSG). Over 2015-2020, the systematic comparison between SSS from L4 OI SSS and in situ measurement reveals an overall RSMD of about 0.75, a global bias of 0.07 and correlation coefficient around 0.92 (Table 1).

MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 statistics have been compared with those computed from scientific version of the product distributed at CATDS: https://doi.org/10.17882/73142 and documented in Kolodziejczyk et al. (2021) (https://doi.org/10.1175/JTECH-D-20-0093.1). They show comparable performance over the period 2015-2017 (SMOS+SMAP).

I.3 Estimated Accuracy Numbers

Table 1: Main Estimated Accuracy numbers

Variable	Units	Metrics	Values
salinity	unitless	Global mean bias, standard deviation, correlation over 2011-2020	0.75, 0.07, 0.92

II PRODUCTION SYSTEM DESCRIPTION

II.1 MULTIOBS SSS/SSD/SSSp processing

Table 2: MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 product specification

Product Specification Customer Name	MULTIOBS_GLO_PHY_SSS_L4_MY_015_015
Geographical coverage	Global: 90°N-S/180°W-180°E
Variables	Sea Surface Salinity, Sea Surface density, Sea Surface temperature, Sea Surface Spiciness, Sea Surface Absolute Salinity, Sea Surface Conservative temperature, Sea Surface thermal expansion coefficient, Sea surface haline contraction coefficient, Sea Ice Mask, SMOS Sea Surface Salinity correction, ISAS Sea Surface Salinity (5 m depth)
Analysis	Yes
Forecast	No
Available time series	Dataset-sss-ssd-sssp: 2011-2022
Temporal resolution	Weekly
Delivery mechanism	CMEMS Information Service
Production Center	CATDS
Horizontal resolution	0.25° on EASE grid (degraded latitudinal resolution at high latitude)
Number of vertical levels	1 level, surface (ISAS bias correction from in situ measurement at 5 m depth)
Format	Netcdf 4.0 CF 1.6

II.2 Short description

Ten years of L-Band radiometric measurements have proven the capability of satellite Sea Surface Salinity (SSS) to resolve large scale to mesoscale SSS features in tropical to subtropical ocean. In mid to high latitude, L-Band measurements still suffer from large scale and time systematic errors. Here, a simple method is proposed to mitigate the large scale and seasonal varying biases. First, in order to estimate the systematic errors, an Optimal Interpolation (OI) using a large correlation scale (~500 km) is used to map ESA Soil Moisture, Ocean Salinity (SMOS) Level 3 products, and NASA Soil Moisture Active Passive (SMAP) Level 3 product (from Remote Sensing System) and is compared to the equivalent mapping of *in situ* observations (In Situ Analysis System, ISAS). Then, a second mapping is performed on adjusted SSS at the scale of SMOS spatial resolution (~45 km). Using the thermodynamic equation of state (TEOS-10), the resulting L4 SSS satellite product is combined with satellite-microwave SST products to estimate sea surface density, spiciness, haline contraction and thermal expansion coefficients, useful parameters to fully characterize the surface ocean water masses. The new L4 SSS products are validated against independent *in situ* measurements mainly from available ship's thermosalinograph (not interpolated in ISAS products).

II.3 Data

SMOS L3 data

The ESA SMOS data are used to retrieve SSS since January 2010 and is still operating. The SMOS satellite is equipped with the Microwave Imaging Radiometer using Aperture Synthesis (MIRAS) instrument which is a 2-dimensional interferometric passive radiometer operated at L-band frequency (1.4 GHz). This synthetic aperture antenna technology provides an average ground spatial resolution on the order of 45 km (between 35 km and 100 km). The SMOS satellite is on a sun-synchronous circular orbit with a local equator-crossing time at 6 AM on the ascending node with a revisit time of 3-5 days (see *Boutin et al.*, 2018).

The SMOS L3 maps used in this study are the 9 day debias L3Q maps produced by the Centre Aval de Treatement des Données SMOS (CATDS) (hereafter L3Q) and mapped on an ESA Equal-Area Scalable Earth (EASE) horizontal grid with a grid resolution of 25 km*25 km. These L3Q maps result from a successive filtering and smoothing with a 9-days Gaussian window of the level 2 retrieved SSS using a Bayesian approach. They are corrected from systematic land-sea contamination, across-swath SSS systematic errors and seasonal latitudinal systematic errors using SMOS SSS self-consistency criteria. The long term SSS absolute calibration is derived from the 80% upper quantile of the ISAS (*In Situ* Analyses System, *in situ* data climatology) 6-year SSS statistical distribution, but the temporal variability of these L3Q SSS is independent of *in situ* SSS. More details about the methodology can be found in *Kolodziejczyk et al.(*2016) and *Boutin et al.*(2018).

SMAP product

The NASA Soil Moisture Active Passive (SMAP) satellite was launched on 31 January 2015 and provides SSS observations since April 2015 from a passive L-band radiometer initially designed to measure soil moisture from space (Fig. 1). The SMAP radiometer allows a ground resolution on the order of 40 km. The SMAP satellite is in a sun-synchronous near-polar orbit at an inclination of 988 and an altitude of 685 km, with an ascending node time of 1800 LT. It completes global coverage in approximately 3 days, with an exact repeat cycle of 8 days (Meissner et al., 2016).

We use the SMAP level-3 v5.3, 8-day, 40-km products provided by Remote Sensing Systems (RSS) and retrieved directly from the geophysical model (Meissner et al., 2018). Over the open ocean, the v5.3

release reduces spurious temporal and zonal biases using the Scripps Argo optimal interpolation (OI) product (Meissner et al., 2018).

The major changes in Version 5.0 from V4.0 are an improved sea-ice detection and correction algorithm and the inclusion of formal uncertainty estimates. The major changes in Version 5.3 when compared to Version 5.0 is removal of salty biases that "haloed" land masses during the early part of the SMAP mission (April 2015 – August 2015).

ISAS products

The In Situ Analysis System (ISAS) provides interpolated fields from *in situ* SSS. Monthly gridded fields of salinity derived from in situ measurements are obtained using an Optimal Interpolation (OI) (*Bretherton et al.*, 1976) tool developed for the synthesis of the Argo global dataset (*Gaillard et al.*, 2016). The interpolation is mainly based on delayed mode Argo floats temperature and salinity measurements. Since 2000's, the International Argo program has allowed the deployment of a global array of autonomous floats that provides quality controlled profiles every 10 days with a quasi-homogeneous global coverage of 3°x3° over the first 2000 meters in depth (*Roemmich et al.*, 2019). Other quality controlled dataset are interpolated with ISAS tool: the Global Tropical Moored Buoy Array (www.pmel.noaa.gov/gtmba/) temperature and salinity timeseries; and CTD from Marine Mammals equipped by French SO MEMO in the framework of Marine Mammals Exploring Ocean Pole to Pole (www.meop.net/groups/france.html).

For the L4 OI product, we use the ISAS-NRT (ISAS in Near-Real Time) monthly fields produced by Coriolis data center in the Framework of Copernicus In Situ Thematic Assembly Center. (www.coriolis.eu.org/Data-Products/Data-Delivery/Copernicus-In-Situ-TAC).

Microwave SST fields

In order to compute density, spiciness, thermal expansion and haline contraction, SST products have been combined with L4 OI SSS data. For the purpose of merging data with the produced L4 OI SSS data, only products with comparable resolution have been used. Therefore, AMSR-E/2 microwave satellite SST products, with comparable spatial resolution on the order of 0.25°*0.25°, have been used for computing L4 parameters fields. AMSR-E was launched on the NASA's EOS Aqua spacecraft, on May 4, 2002 until October 4, 2011.

AMSR-2 was launched on the JAXA's GCOM-W1 spacecraft, on May 18, 2012 and is still operating. AMSR-E/2 SST products are provided by RSS. The AMSR series of instruments are passive microwave radiometers placed in a near-polar orbit which allows for up to twice-daily sampling of a given Earth location. The AMSR instrument has eight microwave channels for measuring SST (skin temperature), wind speed, cloud liquid water, vapor content and rain rate. The SST products used in this study is the weekly product binned on a 0.25° resolution grid, which spans the period from 18 May 2012 until now (*Wentz et al.*, 2014).

II.4 Method

NB: *Here an overview of the method is provided. The method is fully detailed in Kolodziejczyk et al. (2021).*

Objective analysis

Optimal interpolation (OI) of satellite L3 SSS data is performed following *Bretherton et al.* (1976). The interpolated SSS fields consist of the sum of a reference field (or first guess) and an anomaly field obtained by linear combination of satellite SSS data as follows:

$$x^a = x^f + K^{OI}(y^o - y^f) \qquad \textit{Equation 1}$$

where:

$$K^{\mathrm{OI}} = \mathrm{C_{ao}}(C_{\mathrm{oo}} + \mathrm{R})^{-1} \qquad \textit{Equation 2}$$

where x^a are the vectors of the estimates at the grid point and y^o is the vector of observation, respectively; and x^f and y^f are the vectors of the reference field values at the grid and observation points, respectively. $\mathbf{K^{OI}}$ is the weight's matrix constructed from co-variance and error matrix. $\mathbf{C}_{ao}$is the covariance matrix between the analyzed fields at the grid and observation points; $\mathbf{C}_{oo}$is the covariance matrix between the observations; and **R** is the observation error matrix.

The SMOS and SMAP SSS fields are mapped on an EASE horizontal grid with a resolution of about 0.25°. The optimal analysis is calculated on anomalies relative to the reference field used as a first guess i.e. ISAS. The analysis modifies the first guess according to the observations a priori statistics and chosen scales and covariance model. Thus, in poorly sampled areas or if the observational errors are relatively large, the solution may remain close to the reference field.

The covariance matrices $\mathbf{C}_{oo}$and $\mathbf{C}_{ao}$ are constructed using the same covariance model **C**(dr,dt), using a Gaussian function of time and space :

$$C_{\mathrm{i,j}}(\mathrm{dr,dt}) = \sigma_{\mathrm{i,j}} e^{-\left(\frac{\mathrm{dr}_{\mathrm{i,j}}^2}{2L^2} + \frac{\mathrm{dt}_{\mathrm{i,j}}^2}{2T^2}\right)} \qquad \textit{Equation 3}$$

where *dr* and *dt* are the spatial and temporal distances, and *L* and *T* are the associated decorrelation scales. The weight given at each scale of variability corresponds to the variance. The variance field, $\sigma_{\mathrm{i,j}}$, has been computed from filtered historical SMOS data (*Boutin et al.*, 2018). The error matrix for SMOS and SMAP L3 data is provided along with the L3 fields. Optimal interpolation is used to estimate the bias, re-project and filter out the corrected data onto the EASE grid.

Correction of seasonal and large scale errors

The first step consists in a first OI using Equation 1. SMOS SSS L3Q and SMAP SSS L3 v5.3 data are mapped over a period of 7 days separately for each satellite using a 'large' correlation scale comparable to that used in ISAS on the order of 500 km using Equation 3 introduced into the Gaussian correlation model and using the ISAS monthly fields as first guess (*Gaillard et al.*, 2016). This correlation scale has been chosen empirically since the sampling strategy and the scale chosen for ISAS allows to resolve on average a 500 km horizontal feature. Smoothed seasonal and large scale biases are then deduced from the comparison of the SMOS and SMAP smoothed field with ISAS SSS monthly fields interpolated for the corresponding analyses week. Then, large scale and seasonal corrections are derived for each weekly SMOS and SMAP L3 fields. Note that, by construction the ISAS SSS fields are relaxed to climatology in

regions where no in situ data are available (especially at high latitude). It is likely that in these regions the large scale bias correction is less effective.

In the subtropical and tropical regions, SMOS and SMAP L3 large scale and seasonal biases are reduced (*Boutin et al.*, 2018) to values lower than 0.3 pss locally. This is on the order of the expected error on SMOS gridded data (*Boutin et al.*, 2018). Thus, seasonal and large scale corrections are relaxed within the tropics using a Gaussian function centered at this equator:

$$\alpha(y) = 1 - e^{\frac{y^2}{l^2}}$$ *Equation 4*

where the length scale l= 30°. This relaxation allows to keep the information content in the data of the low salinity plume of large tropical river runoff as well as the Tropical Instability Wave signature in SSS found in SMOS products. The large scale correction is also relaxed in the Gulf of Mexico and off Rio de la Plata river mouth.

Noise reduction and data mapping

Finally, the corrected SMOS and SMAP L3 SSS are mapped using an interpolation scheme by introducing into the Gaussian function (Equation 3) spatial and temporal correlation scales corresponding to the synoptic scales resolved by SMOS, i.e. 7 days and 25 km.

III VALIDATION FRAMEWORK

III.1 Validation dataset

Between 2015 and 2020, near surface *in situ* salinity and temperature observations are provided by thermosalinographs (TSG) installed on voluntary merchant ships (*Alory et al.*, 2015). They provide solid SSS estimates representative of the upper 10 meters with a ~2.5 km horizontal resolution along the ship track. TSG data are independent from ISAS maps since they are not interpolated along with other *in situ* datasets in ISAS fields, except in the ISAS-NRT produced at Coriolis data center. However, correlations scales used in ISAS-NRT smooth out the small and mesoscale features from TSG measurements, thus the independent validation of these scales remain possible. Noise on individual ship SSS data is estimated to be on the order of 0.08 on Practical Salinity Scale of 1978 (here after pss) (*Alory et al.*, 2015). These TSG dataset samples mostly the low to mid latitudes, up to 60°N-S.

III.2 Validation method and metrics

Collocation: Weakly L4 OI SSS satellite products have been co-located with available independent *in situ* observations mainly provided by TSG data from research vessels and ship of opportunity (2 408 887 co-localisations). The collocations are performed pairing each TSG measurement with the occupied satellite 0.25°*0.25° pixel and within the week corresponding to the L4 map. L4 OI SSS SMOS+SMAP, OI L4 SSS SMOS ONLY and ISAS products are compared to in situ data (named satellite L4 OI, L4 OI SMOS ONLY and ISAS, respectively for the period 2015 to 2020).

Metrics: *Differences distribution, RMSD, bias and Correlation for L4 OI SMOS+SMAP (output data), L4 OI SMOS ONLY (output data) and ISAS background (input data), collocation map.*

IV VALIDATION RESULTS

Over 2015-2020, the systematic comparison between SSS from L4 OI SSS and in situ measurement reveals an overall RSMD of about 0.75, a global bias of 0.07 and correlation coefficient around 0.92 (Table 3). These statistics are comparable to the research L4 OI products provided at CATDS (Kolodziejczyk et al., 2021). The slight differences may result from: i) the statistics are not computed over the same period (2011-2014 for the period including SMOS alone in Kolodziejczyk et al., 2021); ii) The input data are not exactly the same, i.e. CATDS L3Q products and ISAS-NRT instead of SMOS L3 CEC debias and ISAS15 products, respectively, in Kolodziejczyk et al., (2021).

Over the period 2015-2020, the statistics have been computed for individual years in order to show the consistency of global statistics over the period: the L4 OI SSS RMSD range between 0.58 (2020) and 0.94 (2018); bias between 0.05 and 0.13; correlation between 0.96 and 0.85.

This reveals a relatively good consistency of the L4 OI SSS quality year after year. Interestingly, L4 OI SSS SMOS+SMAP and L4 OI SSS SMOS ONLY show very comparable statistics over the period 2015-2020. Individual years reveal slightly better performance for the L4 OI SMOS+SMAP product such as 2015, 2019, 2020, while 2016, 2017, 2018 show slightly better performance for L4 OI SMOS ONLY product.

Table 3: Global statistics from co-localisation between in situ data TSG L4 OI SSS, OI SSS SMOS ONLY (output dataset) and ISAS SSS products (input dataset): Root Mean Squared difference (RMSD); Bias; Correlation and Number of co-localisations.

Year	Product	RMSD	Bias	Corr.	N
2015	OI SMOS SMAP	0.66	0.05	0.96	330360
	OI SMOS ONLY	0.67	0.05	0.96	330360
	ISAS	0.79	0.10	0.95	597280
2016	OI SMOS SMAP	0.75	0.07	0.95	387452
	OI SMOS ONLY	0.72	0.02	0.95	387452
	ISAS	0.85	0.11	0.94	543698
2017	OI SMOS SMAP	0.84	0.03	0.92	379514
	OI SMOS ONLY	0.77	0.05	0.92	379514
	ISAS	1.06	0.05	0.87	380833
2018	OI SMOS SMAP	0.94	0.13	0.92	457311
	OI SMOS ONLY	0.69	0.05	0.94	457311
	ISAS	1.36	0.21	0.82	459251
2019	OI SMOS SMAP	0.70	0.08	0.94	431662
	OI SMOS ONLY	0.76	0.08	0.93	431662
	ISAS	0.87	0.16	0.91	600193
2020	OI SMOS SMAP	0.58	0.06	0.85	422588
	OI SMOS ONLY	0.70	0.06	0.87	422588
	ISAS	0.72	0.01	0.82	426699
Total	OI SMOS SMAP	0.75	0.07	0.92	2408887
	OI SMOS ONLY	0.72	0.07	0.93	2408887
	ISAS	0.95	0.11	0.89	3007954

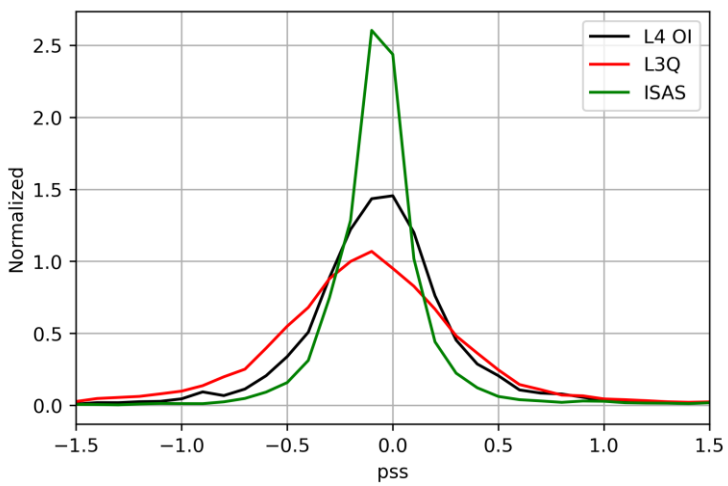

Figure 1: 2017 example of distribution of collocated differences between in situ TSG SSS and L4 OI (black), L4 OI SMOS ONLY (red) and ISAS (green) products.

Eventually, the spatial differences between in situ SSS and L4 OI, L4 OI SMOS ONLY and ISAS SSS are analized for 2017 (Figure 2). L4 OI SSS (Figure 2ab) exhibits an overall agreement with in-situ data with largest difference in near coastal region (largest representativeness error with TSG data and RFI[1]). Analysis of the 2015 and 2020 period provides comparable results (not shown).

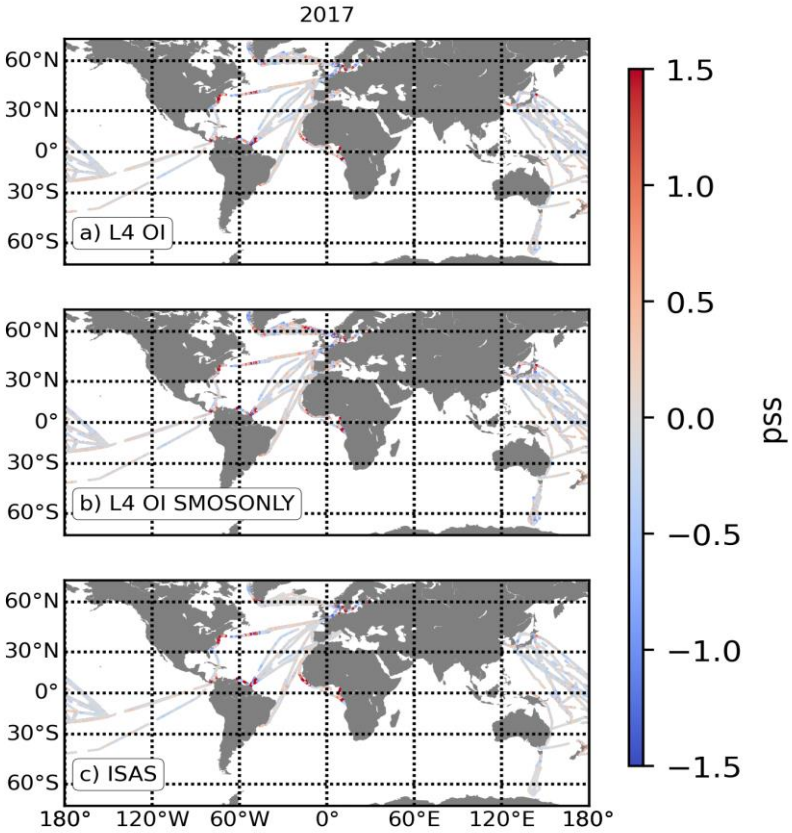

Figure 2: 2017 example of spatial distribution of collocated differences between in situ TSG SSS and L4 OI (a), L4 OI SMOS ONLY (b) and ISAS (c) products.

1 Radio frequency interference.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version	Other
June 2024	Extension of the series to SMOS+SMAP data		

VI QUALITY CHANGES SINCE PREVIOUS VERSION

No significant quality change has been observed from previous version.

VII REFERENCES

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